Investment Comparator

COM-480 Data Visualization — Milestone 2 Report

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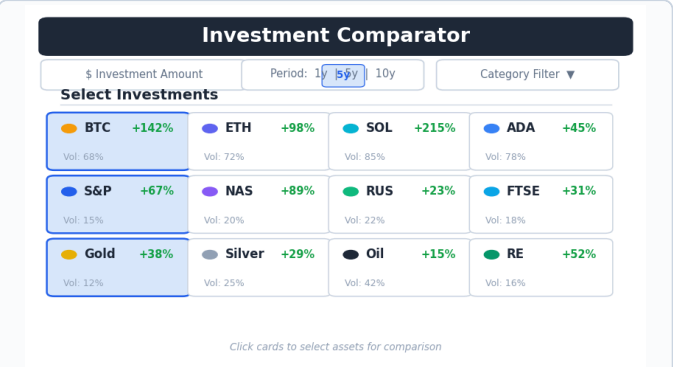
PROJECT GOAL

Our project is an **interactive investment comparison tool** that enables users to explore and contrast the historical performance of assets across three major categories: cryptocurrencies (Bitcoin, Ethereum, Solana, Cardano, Polygon, Chainlink, Avalanche), traditional markets (S&P 500, NASDAQ, Russell 2000, FTSE 100, Nikkei 225, DAX, QQQ ETF, VTI ETF, US Real Estate), and commodities (Gold, Silver, WTI Oil, Copper, Natural Gas). Built on 11 years of real historical data (2014–2024), the tool lets users select any combination of 20 assets, adjust time horizons (1, 5, or 10 years), and set a custom initial investment to see exactly how their portfolio would have performed.

The core motivation is to **democratize financial analysis**. Tools like Portfolio Visualizer or Bloomberg Terminal are either expensive or intimidating for non-professionals. Our target audience—university students, beginner investors, and finance enthusiasts—deserves an accessible, visually engaging alternative that turns raw data into intuitive insights. Through interactive time-series charts, side-by-side result cards with risk badges, a Dollar-Cost Averaging (DCA) simulator, and a risk analysis dashboard, users can answer the fundamental question: *"Is crypto worth the risk compared to gold, stocks, and real estate?"*

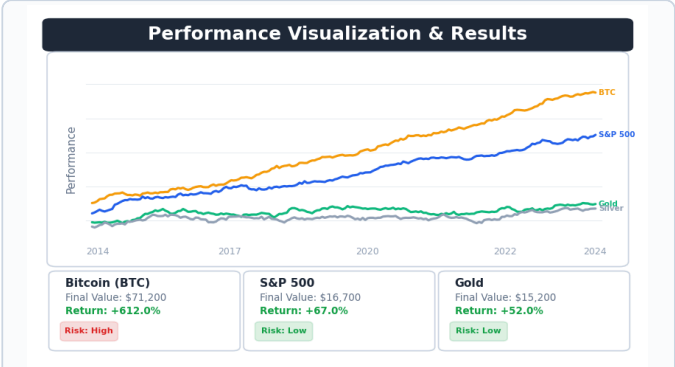
Beyond simple price comparison, the tool provides educational value: volatility indicators contextualize risk, DCA simulations demonstrate disciplined investing strategies, and risk metrics (Sharpe Ratio, Maximum Drawdown) offer professional-grade insight in an approachable format. The final version will expand to 15+ risk indicators, correlation matrices, and portfolio-building features.

WIREFRAMES & INTERFACE DESIGN



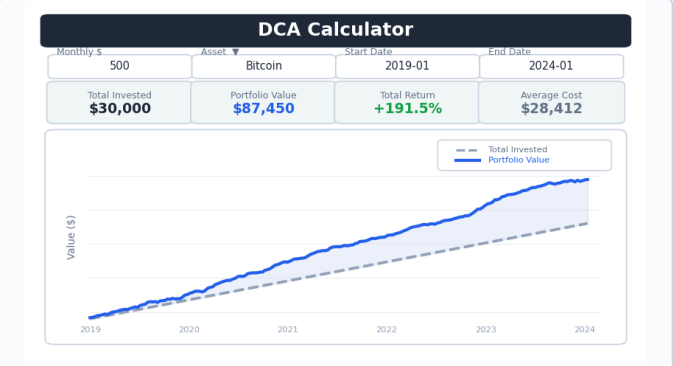
Main Comparator — Asset Selection

Category-organized grid with investment amount, period, and filter controls



Performance Visualization & Results

Multi-line normalized chart with per-asset result cards and risk badges



DCA Calculator

Monthly investment simulation: invested vs. portfolio value over time

Risk Analysis Dashboard

Volatility, Sharpe Ratio, and Max Drawdown comparison across assets

TOOLS & LECTURE CONNECTIONS

VISUALIZATION	TOOL / LIBRARY	RELATED LECTURE
Interactive asset selection & filtering	React 19 + TypeScript	**L5: Interactions** (filtering, brushing, linked views, Shneiderman's mantra)
Multi-line time series charts	Recharts (LineChart)	**L4: D3.js** (d3.line() path generator) + **L7: Do and dont in viz** (line chart best practices)
Performance comparison cards	Next.js 16 + Tailwind CSS	**L7: Designing viz** (task abstraction, design process) + **L11: Tabular data** (LineUp rankings)
Risk metrics dashboard	Custom TypeScript calculations	**L6: Mark, channels** (effectiveness rankings, visual encoding) + **L4: Data** (Stevens' scales)
DCA investment simulation	JavaScript + Recharts	**L12: Storytelling** (Freytag's pyramid, martini glass structure, progressive reveal)
Color-coded categories & risk badges	Tailwind CSS v4	**L6: Perception colors** (categorical/sequential scales, ColorBrewer, preattentive processing)

- **Data sources:** CoinMarketCap, Yahoo Finance, EIA, Macrotrends, Kitco, FRED (Case-Shiller)
- **Deployment:** Vercel (planned) | **EDA:** Python, pandas, matplotlib, seaborn

IMPLEMENTATION BREAKDOWN

DONE Core Visualization (MVP)

- Interactive 20-asset selection by category
- Time period controls (1y / 5y / 10y)
- Multi-line normalized performance chart
- Side-by-side comparison cards with risk badges
- Category performance analysis grid
- Detailed comparison table with category tags
- DCA calculator (6 assets, performance chart)
- Basic risk analysis (Volatility, Sharpe, Max DD)
- Responsive design (desktop & mobile)

IN PROGRESS Teaser UI & Previews

- Portfolio Builder interface (Coming Soon badge)
- Static correlation matrix preview from EDA
- Historical Events toggle (disabled, labeled)
- DCA dropdown with 21 assets (6 functional)
- Advanced Metrics preview modal (15+ listed)
- PDF export button (Beta badge)

PLANNED Advanced Risk Metrics

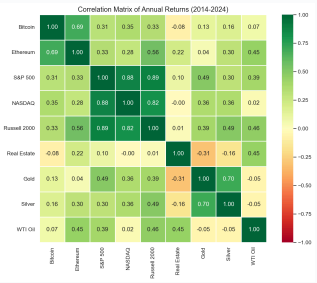
- 15+ professional indicators: Sortino, VaR, Calmar, Beta, Alpha, Treynor, Information Ratio
- Interactive risk-return scatter plot
- Dynamic correlation heatmap with filtering
- Historical event annotations on charts

PLANNED Portfolio & Analytics

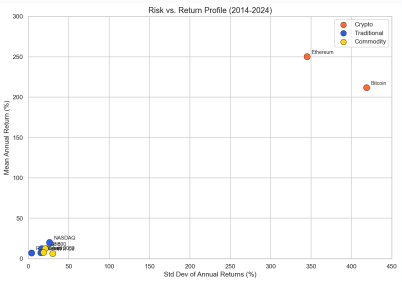
- Markowitz portfolio optimizer with efficient frontier
- Diversification insights & rebalancing suggestions
- Expand DCA to full 21-asset analysis
- PDF export of analysis results
- Tooltip explanations for financial terms
- Guided tour for beginner users

EXPLORATORY DATA ANALYSIS

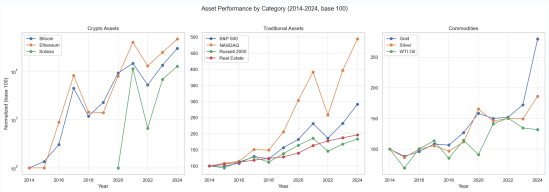
Key findings from our EDA (full analysis in `eda.ipynb`): crypto assets are extreme outliers in both return and volatility, stock indices cluster tightly (r = 0.82–0.89), and Gold/Silver form a distinct hedging group (r = 0.70). Real Estate shows the lowest volatility (std 3.8%) with steady positive returns.



Correlation Matrix (2014–2024)
Cross-asset return correlations reveal three distinct clusters



Risk vs. Return Profile
Crypto: extreme risk-return outliers vs. traditional assets



Asset Performance by Category
Normalized growth (base 100) across crypto, stocks, commodities

Repository: github.com/com-480-data-visualization/-burek- • **Stack:** Next.js 16 · React 19 · TypeScript · Recharts · Tailwind CSS v4 • **Status:** MVP functional, advanced features in development for final presentation